

Dhruv Patel

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EDUCATION

University of Houston

Houston, TX

M.S. Computer Science, GPA: 3.56/4.0

2025 – 2027

Coursework: Visualization, Data Mining, Advance Numerical Analysis, Cloud Computing

Gujarat Technological University

India

B.E. Computer Engineering, CGPA: 8.76/10

2025

Coursework: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks

EXPERIENCE

Software Developer Intern

Jan 2025 – Jun 2025

Narola Infotech

Remote

- Engineered full-stack Android platform using Kotlin, Firebase, and REST APIs serving real-time users.
- Designed scalable MVVM architecture and modular state management improving feature velocity.
- Built secure authentication and cloud-backed data pipelines handling dynamic content delivery.
- Reduced location errors by 25% after migrating to Google Maps infrastructure.

PROJECTS

Autonomous AI Agent Platform (RAG + Tool Calling + Memory)

- Developed autonomous AI agent capable of multi-step reasoning, tool usage, and contextual memory.
- Integrated tool-calling for web search, document analysis, and structured output generation.
- Designed retrieval + memory pipeline using embeddings and vector storage.
- Built modular FastAPI backend and containerized scalable execution services.
- Tech:** Python, LLM APIs, Vector DB, FastAPI, Docker

Scalable Distributed Task Processing Queue Management System

- Built distributed task queue system handling asynchronous background jobs and batch workflows.
- Implemented worker orchestration, retry logic, and monitoring dashboard.
- Designed REST job submission APIs and scalable execution pipeline using Redis queues.
- Tech:** Python, Redis, FastAPI, Docker

AI Document Intelligence System — Domain Analyzer + Smart Summary

- Built AI document analyzer generating domain-specific summaries and detecting document categories.
- Designed prompt pipelines for summarization, insight extraction, and classification.
- Achieved sub-2s response latency using optimized LLM workflows.
- Exposed backend APIs via FastAPI and containerized using Docker.
- Tech:** Python, OpenAI API, FastAPI, Docker

Swad Seeker — Production Android Platform

- Developed scalable Android application with authentication, maps, and real-time cloud sync.
- Integrated REST APIs and Firebase backend for dynamic data delivery.
- Designed modular architecture enabling future AI-based recommendation features.

TECHNICAL SKILLS

Programming: Python, Kotlin, Java, C/C++, SQL, JavaScript

Backend & APIs: FastAPI, RESTful APIs, API Design, Microservices, Authentication (JWT/OAuth), Firebase, Redis, Server-side Development

Databases: MySQL, PostgreSQL, SQLite, Firebase Firestore, Database Design, Query Optimization

AI/LLM: LLM APIs, RAG Pipelines, Prompt Engineering, Vector Databases, Embeddings, Semantic Search, OpenAI API

Cloud & DevOps: Docker, Linux/Unix, CI/CD Basics, Cloud Deployment

Tools: Git, GitHub, Postman, VS Code, Android Studio

Core CS: Data Structures & Algorithms, OOP, DBMS, System Design, Multithreading, Distributed Systems